Topic name

```
/environment
```

Message type

```
float32 xdot_ms         # m/s
float32 ydot_ms         # m/s
float32 salinity_ppt    # parts per thousand
float32 temperature_c   # celsius
```

Publishing frequency

Every 10 milliseconds.

Configuration

These configuration keys should be added to the environment_config file.

```
"env_data_source": "NOAAGRTOFS"
```

Options are:

- NOAAGRTOFS uses the NOAA Global Real-Time Ocean Forecast System web API
- REPUBLISH re-publishes data already present on other topics
- CACHE uses cached .csv data (see **env_data_cache**)
- STATIC uses static values (see **env_data_static**)

```
"env_data_cache": "src/databases/environment_data_cache"
```

Path to directory of .csv files that are used to publish data to /environment for a CACHE data source.

```
"env_data_static": {
    "xdot_ms": 0,
    "ydot_ms": 0,
    "salinity_ppt": 35,
    "temperature_c": 15,
}
```

Data to publish for a STATIC data source.

```
"env_data_smoothing": true
```

Data resolution may be limited (eg, periodic API calls, inconsistent .csv data). In this case, the system should implement a smoothing feature between data points as the vehicle navigates.

Behavior

- If a data source is initially available, but becomes unavailable during the simulation, attempt to locate data at the fallback source. If no data is available at the fallback source, use the last-known data. The fallback source for NOAAGRTOFS is CACHE. CACHE has no fallback.
- If a data source is not available from the start of a simulation, fall back to STATIC data.
- Generate a warning every 5 seconds while a fallback data source is used.